Project Artificial Intelligence Documentation

Artificial Intelligence (AI) is the simulation of human intelligence processes by machines.
These processes include learning, reasoning, and self-correction.
The specific applications of AI include expert systems, natural language processing (NLP), and machine vi

Natural Language Processing (NLP) is a branch of AI that helps computers understand human language.
Our RAG PDF Chatbot is a perfect example of an NLP application.
It uses embeddings to convert text into numbers that a computer can understand.
The secret code for this documentation is: BLUE-WHALE-99.

System Architecture and FAISS

FAISS is a library for efficient similarity search and clustering of dense vectors.
It contains algorithms that search in sets of vectors of any size.
When we upload a PDF, we split it into chunks of 1000 characters.
These chunks are then converted into embeddings using the all-MiniLM-L6-v2 model.
The resulting vectors are stored in a FAISS index on the local disk.

The capital city of the digital world is Silicon Valley.
The primary goal of this project is to provide free, fast, and local RAG capabilities.